

SQL Queries along with Solutions

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Query 1:

1

```
SELECT * FROM orders LIMIT 10;
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSo
1	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram
2	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral
3	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FRESHSHIP	Email
4	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook
5	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email
6	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram
7	ORD200006	2025-06-17	C83492	Laptop	1	664.42	956 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook

Execution finished without errors.
Result: 10 rows returned in 10ms
At line 1:
SELECT * FROM orders LIMIT 10;

Query 2:

1
2
3
4
5
6
7

```
-- Just see product and price info  
SELECT Product, UnitPrice, TotalPrice  
FROM orders;  
  
-- Give columns friendlier names (aliases)  
SELECT Product AS ItemName, TotalPrice AS Revenue  
FROM orders;
```

	ItemName	Revenue
1	Monitor	2853.1
2	Phone	302.7
3	Tablet	2753.4
4	Chair	273.19
5	Printer	2504.04
6	Phone	491.72
7	Laptop	664.42

Execution finished without errors.
Result: 1200 rows returned in 20ms
At line 5:
-- Give columns friendlier names (aliases)
SELECT Product AS ItemName, TotalPrice AS Revenue
FROM orders;

Query 3:

1

-- Orders that were shipped

2

SELECT * FROM orders

3

WHERE OrderStatus = 'Shipped';

4

5

-- Big-ticket orders

6

SELECT OrderID, Product, TotalPrice

7

FROM orders

8

WHERE TotalPrice >= 1000;

9

10

-- Orders paid by Debit Card AND shipped (combine conditions)

11

SELECT * FROM orders

12

WHERE PaymentMethod = 'Debit Card' AND OrderStatus = 'Shipped';

13

14

-- Orders that came from Instagram or Referral

15

SELECT * FROM orders

16

WHERE ReferralSource IN ('Instagram', 'Referral');

17

18

-- Pattern matching: addresses on 'Main St'

19

SELECT ShippingAddress FROM orders

20

WHERE ShippingAddress LIKE '%Main St%';

	ShippingAddress
1	928 Main St
2	823 Main St
3	512 Main St
4	275 Main St
5	668 Main St
6	934 Main St
7	986 Main St
8	706 Main St
9	904 Main St
10	102 Main St
11	333 Main St
12	831 Main St

Execution finished without errors.
Result: 1200 rows returned in 8ms
At line 18:
-- Pattern matching: addresses on 'Main St'
SELECT ShippingAddress FROM orders
WHERE ShippingAddress LIKE '%Main St%';

Query 4:

1

-- Highest revenue orders first

2

SELECT OrderID, Product, TotalPrice

3

FROM orders

4

ORDER BY TotalPrice DESC;

5

6

-- Most recent orders first

7

SELECT OrderID, Date

8

FROM orders

9

ORDER BY Date DESC;

	OrderID	Date
1	ORD200256	2025-06-30
2	ORD201107	2025-06-30
3	ORD200176	2025-06-28
4	ORD200773	2025-06-28
5	ORD200882	2025-06-28
6	ORD201039	2025-06-28
7	ORD200228	2025-06-27
8	ORD200064	2025-06-25

Execution finished without errors.
Result: 1200 rows returned in 22ms
At line 6:
-- Most recent orders first
SELECT OrderID, Date
FROM orders
ORDER BY Date DESC;

Query 5:

1

-- How many orders per product?

2

SELECT Product, COUNT(*) AS OrderCount

3

FROM orders

4

GROUP BY Product;

5

-- How many orders per payment method?

6

SELECT PaymentMethod, COUNT(*) AS NumOrders

7

FROM orders

8

GROUP BY PaymentMethod;

9

	PaymentMethod	NumOrders
1	Cash	246
2	Credit Card	234
3	Debit Card	232
4	Gift Card	230
5	Online	258

Execution finished without errors.

Result: 5 rows returned in 20ms

At line 6:

-- How many orders per payment method?

SELECT PaymentMethod, COUNT(*) AS NumOrders

FROM orders

GROUP BY PaymentMethod;

Query 6:

1

-- Total revenue by product

2

SELECT Product, SUM(TotalPrice) AS TotalRevenue

3

FROM orders

4

GROUP BY Product

5

ORDER BY TotalRevenue DESC;

6

-- Average order value by referral source

7

SELECT ReferralSource, AVG(TotalPrice) AS AvgOrderValue

8

FROM orders

9

GROUP BY ReferralSource;

10

-- Count orders AND total revenue AND average price together

11

SELECT PaymentMethod,

12

COUNT(*) AS NumOrders,

13

SUM(TotalPrice) AS Revenue,

14

AVG(TotalPrice) AS AvgOrder

15

FROM orders

16

GROUP BY PaymentMethod

17

ORDER BY Revenue DESC;

18

19

	PaymentMethod	NumOrders	Revenue	AvgOrder
1	Credit Card	234	263847.63	1127.55397435897
2	Online	258	262442.94	1017.22069767442
3	Cash	246	259786.29	1056.04182926829
4	Gift Card	230	246323.92	1070.97356521739
5	Debit Card	232	232361.18	1001.55681034483

Execution finished without errors.

Result: 5 rows returned in 2ms

At line 12:

-- Count orders AND total revenue AND average price together

SELECT PaymentMethod,

COUNT(*) AS NumOrders,

SUM(TotalPrice) AS Revenue,

AVG(TotalPrice) AS AvgOrder

FROM orders

GROUP BY PaymentMethod

ORDER BY Revenue DESC;

Query 7:

```
1  -- Only show products that generated more than $50,000 total
2  SELECT Product, SUM(TotalPrice) AS TotalRevenue
3  FROM orders
4  GROUP BY Product
5  HAVING SUM(TotalPrice) > 50000;
6
7  -- Only payment methods used more than 100 times
8  SELECT PaymentMethod, COUNT(*) AS NumOrders
9  FROM orders
10 GROUP BY PaymentMethod
11 HAVING COUNT(*) > 100;
```

	PaymentMethod	NumOrders
1	Cash	246
2	Credit Card	234
3	Debit Card	232
4	Gift Card	230
5	Online	298

Execution finished without errors.
Result: 5 rows returned in 22ms
At line 7:
-- Only payment methods used more than 100 times
SELECT PaymentMethod, COUNT(*) AS NumOrders
FROM orders
GROUP BY PaymentMethod
HAVING COUNT(*) > 100;

Query 8:

```
1  SELECT Product,
2     COUNT(*) AS TotalOrders,
3     SUM(TotalPrice) AS Revenue,
4     ROUND(AVG(TotalPrice), 2) AS AvgOrderValue
5  FROM orders
6  WHERE OrderStatus = 'Shipped'
7  GROUP BY Product
8  HAVING SUM(TotalPrice) > 10000
9  ORDER BY Revenue DESC;
```

	Product	TotalOrders	Revenue	AvgOrderValue
1	Printer	42	44397.41	1057.08
2	Chair	31	40350.44	1301.63
3	Laptop	29	40237.23	1387.49
4	Tablet	40	34828.15	870.7
5	Desk	33	33424.89	1012.88
6	Monitor	32	30924.82	966.4
7	Phone	28	21996.64	785.59

Execution finished without errors.
Result: 7 rows returned in 10ms
At line 1:
SELECT Product,
COUNT(*) AS TotalOrders,
SUM(TotalPrice) AS Revenue,
ROUND(AVG(TotalPrice), 2) AS AvgOrderValue
FROM orders
WHERE OrderStatus = 'Shipped'
GROUP BY Product
HAVING SUM(TotalPrice) > 10000
ORDER BY Revenue DESC;